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Department of CSE

B.Tech. CSE Trimester -5

2520030442
Aleti Sai Pranathi

Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

Experiment -2

AIM: 2a) Enabling Multi-Factor Authentication to Secure Your Access.

Description:

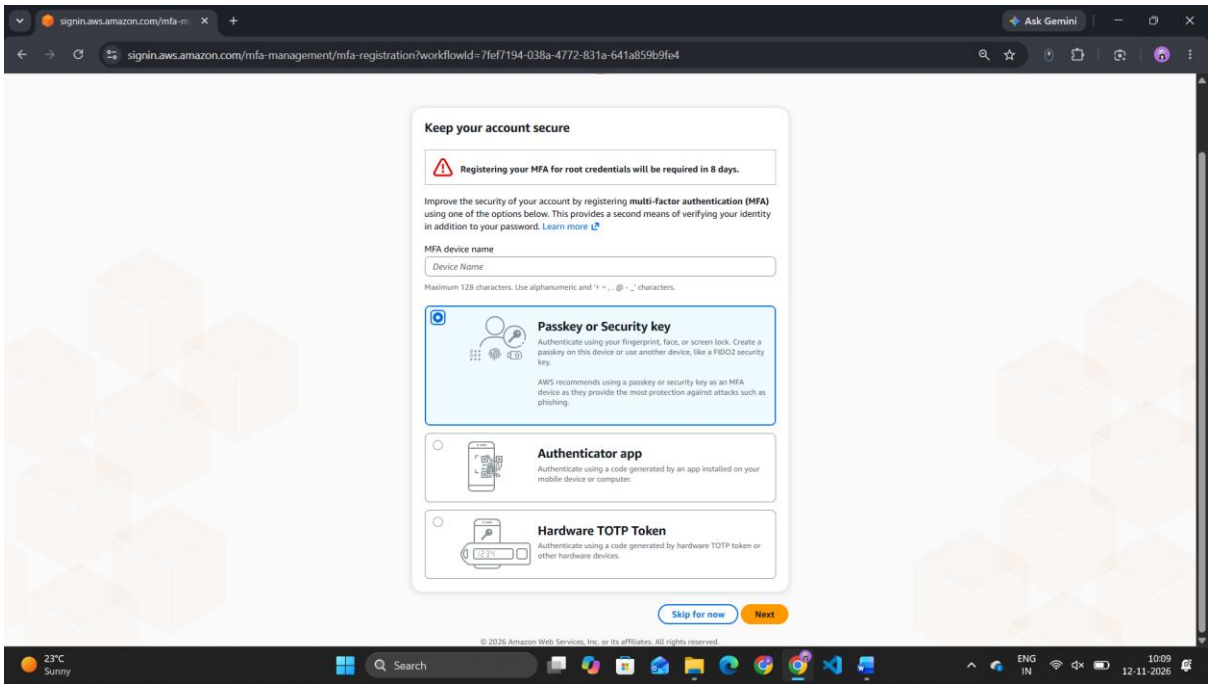
Multi-Factor Authentication (MFA) is a security feature that provides an additional layer of protection to an AWS account. It requires users to verify their identity using a password and a second authentication method, such as an authenticator app or security key.

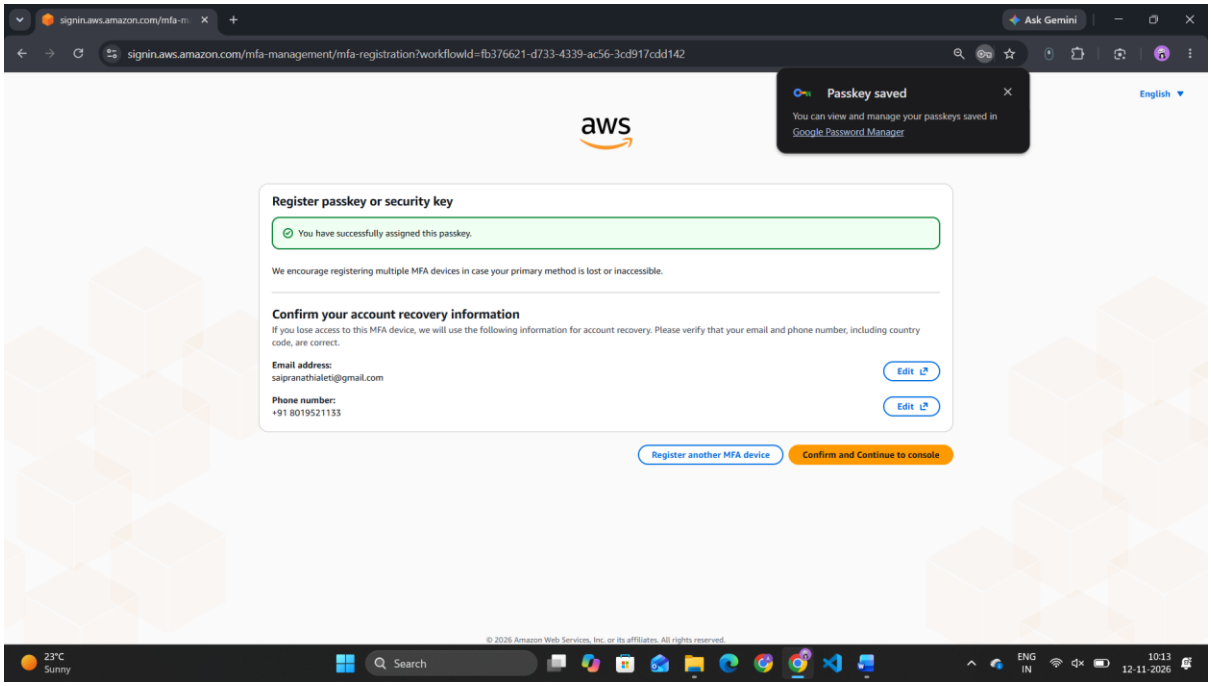
Steps:

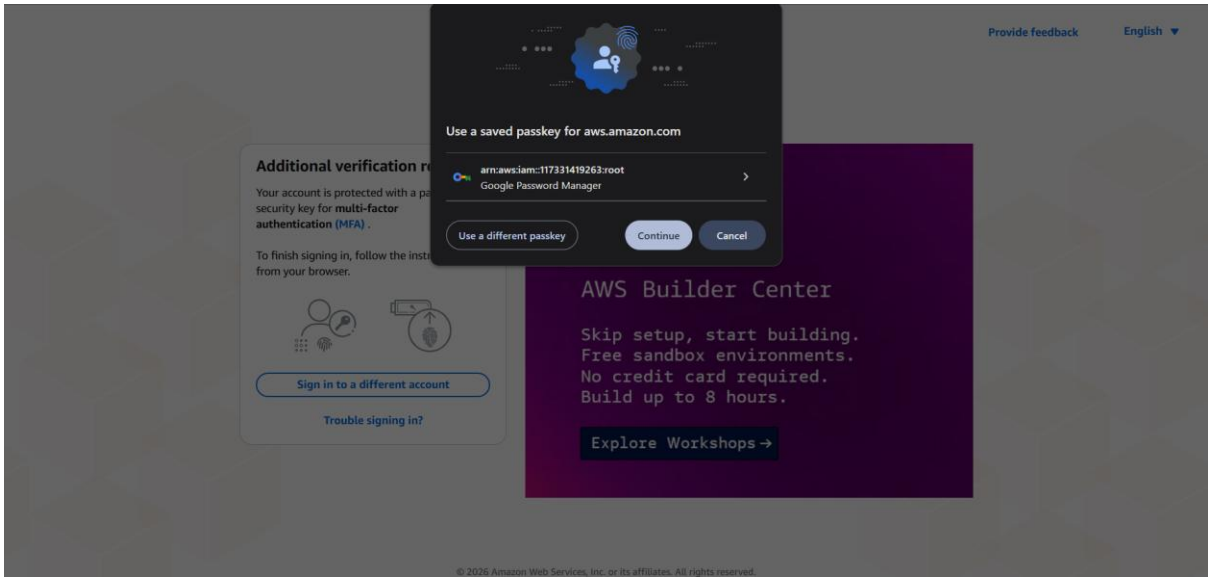
1. Sign in to the AWS Management Console.
2. Open the account menu and select Security Credentials.
3. Navigate to the Multi-Factor Authentication (MFA) section.
4. Click Assign MFA device and enter a device name.
5. Select an authentication method, such as an authenticator app, and scan the QR code.
6. Enter the two consecutive MFA codes generated by the app and click Add MFA.

Output:

2A: Passkey




signin.aws.amazon.com/mfa-management/mfa-registration?workflowId=fb376621-d733-4339-ac56-3cd917cdd142
Ask Gemini
Passkey saved
You can view and manage your passkeys saved in Google Password Manager
English
aws
Register passkey or security key
You have successfully assigned this passkey.
We encourage registering multiple MFA devices in case your primary method is lost or inaccessible.
Confirm your account recovery information
If you lose access to this MFA device, we will use the following information for account recovery. Please verify that your email and phone number, including country code, are correct.
Email address:
saipranathialeti@gmail.com
Edit
Phone number:
+91 8019521133
Edit
Register another MFA device
Confirm and Continue to console
© 2026 Amazon Web Services, Inc. or its affiliates. All rights reserved.
23°C
Sunny
Search
ENG
IN
10:13
12-11-2026


Provide feedback
English
Use a saved passkey for aws.amazon.com
arn:aws:iam::117331419263:root
Google Password Manager
Use a different passkey
Continue
Cancel
Additional verification r
Your account is protected with a pa
security key for multi-factor
authentication (MFA).
To finish signing in, follow the instr
from your browser.
Sign in to a different account
Trouble signing in?
AWS Builder Center
Skip setup, start building.
Free sandbox environments.
No credit card required.
Build up to 8 hours.
Explore Workshops →
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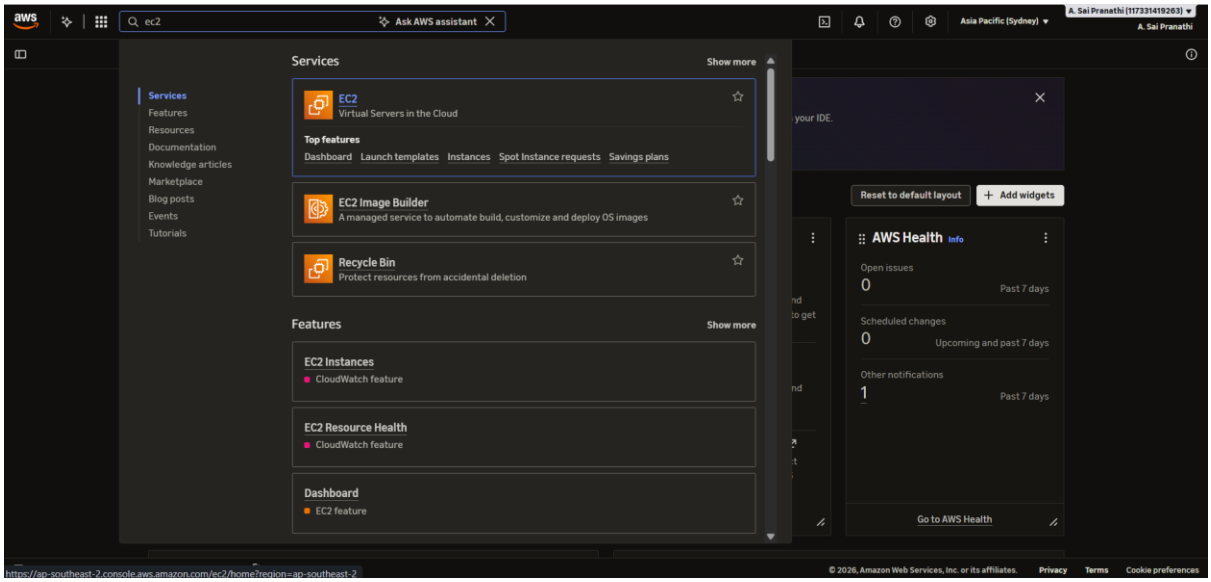
AIM: 2b) Creating First Linux Instance
Description:

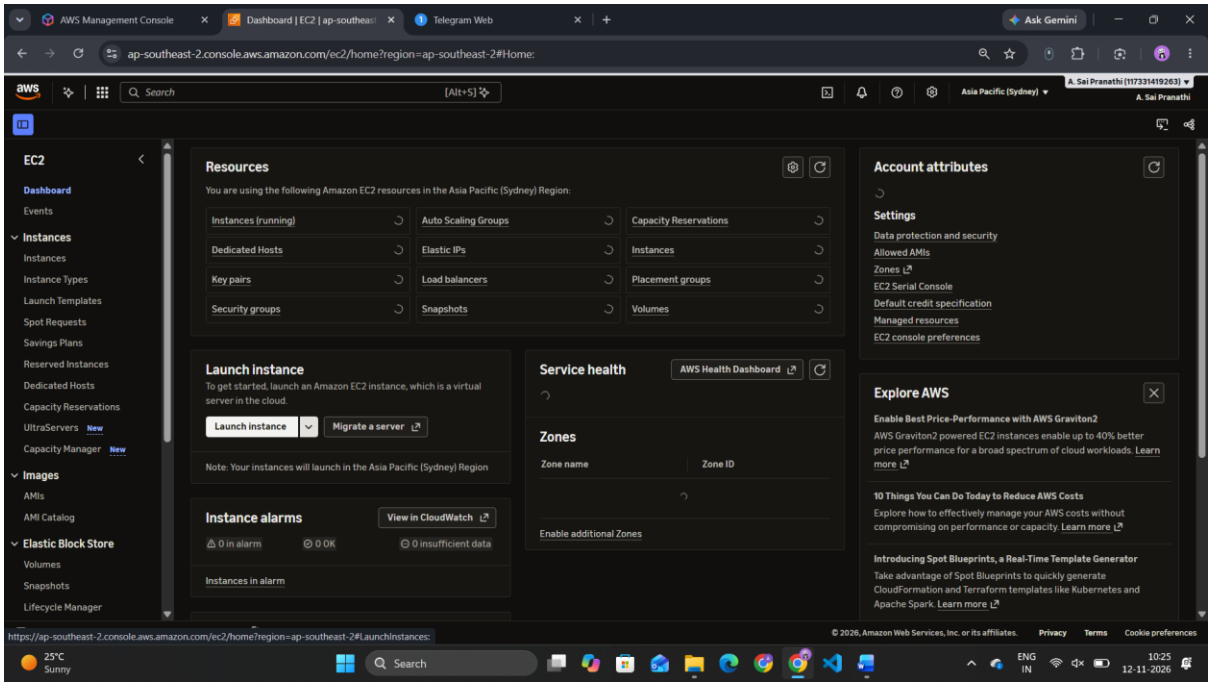
Amazon EC2 (Elastic Compute Cloud) is an AWS service that provides virtual servers called instances. A Linux instance allows users to run Linux-based applications and perform various computing tasks in the cloud.

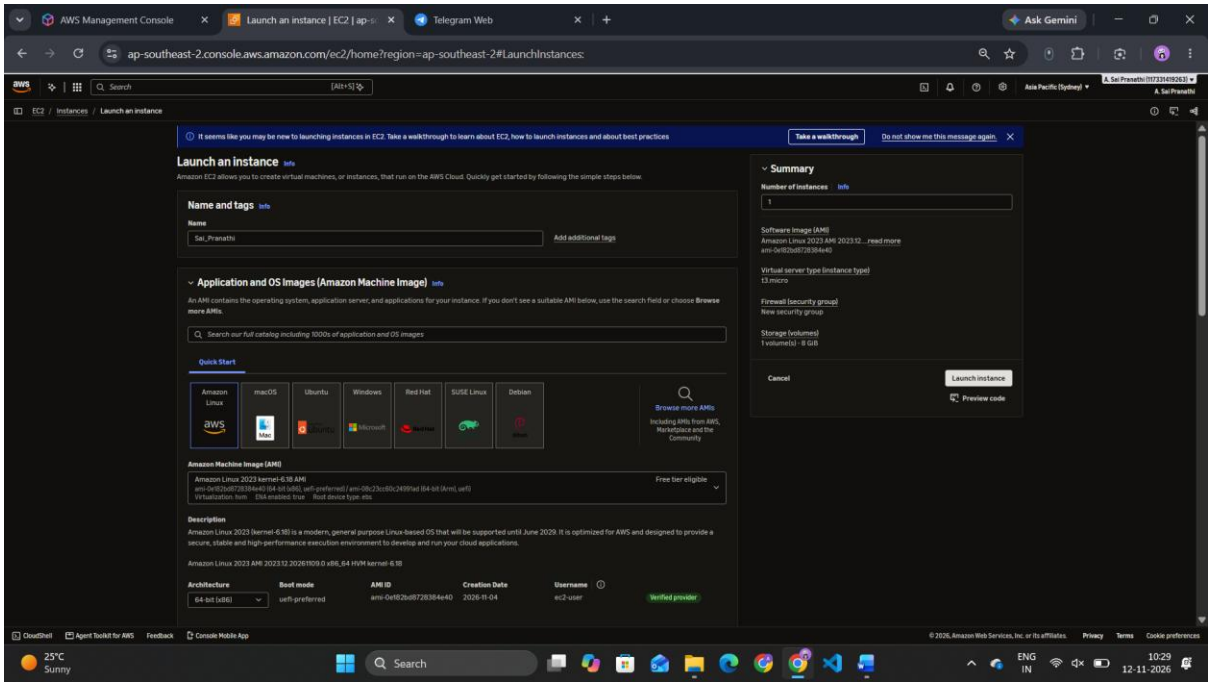
Steps:

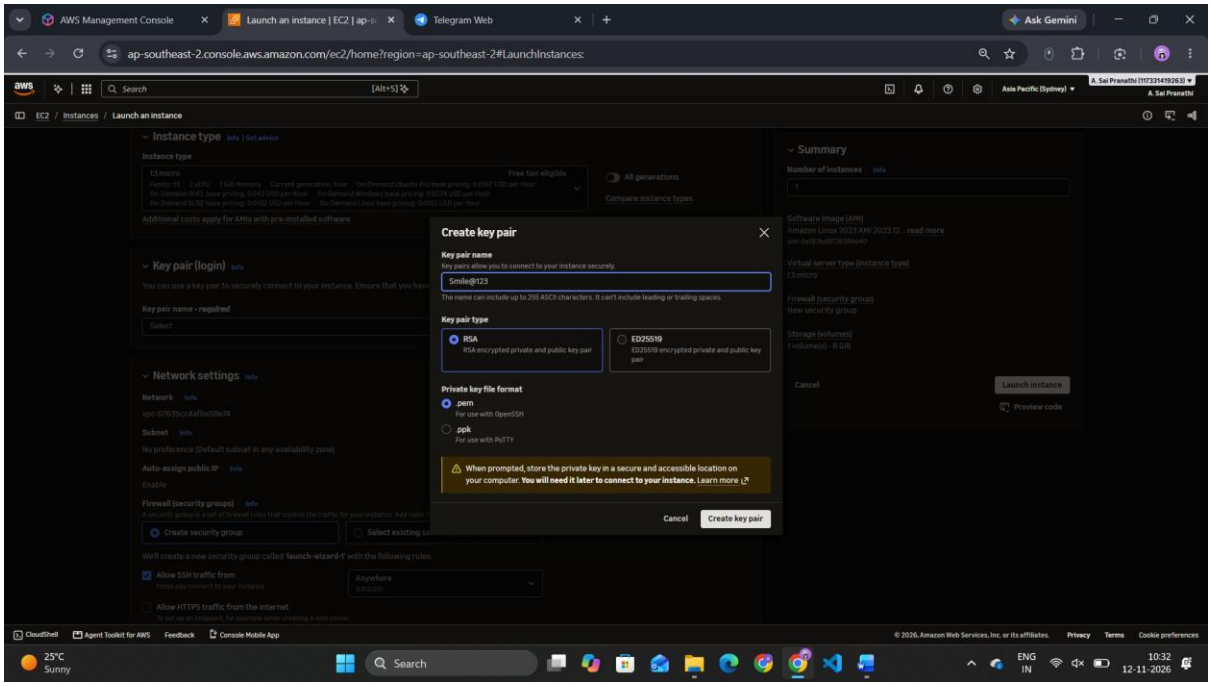
1. Sign in to the AWS Management Console and open the EC2 service.
2. Click on Launch Instance to create a new virtual machine.
3. Enter an instance name and select an Amazon Linux AMI.
4. Choose an instance type, such as t2.micro or an eligible free-tier instance type.
5. Create or select a key pair to securely connect to the instance.
6. Configure network settings and allow SSH traffic (port 22).
7. Click Launch Instance and wait until the instance state changes to Running.

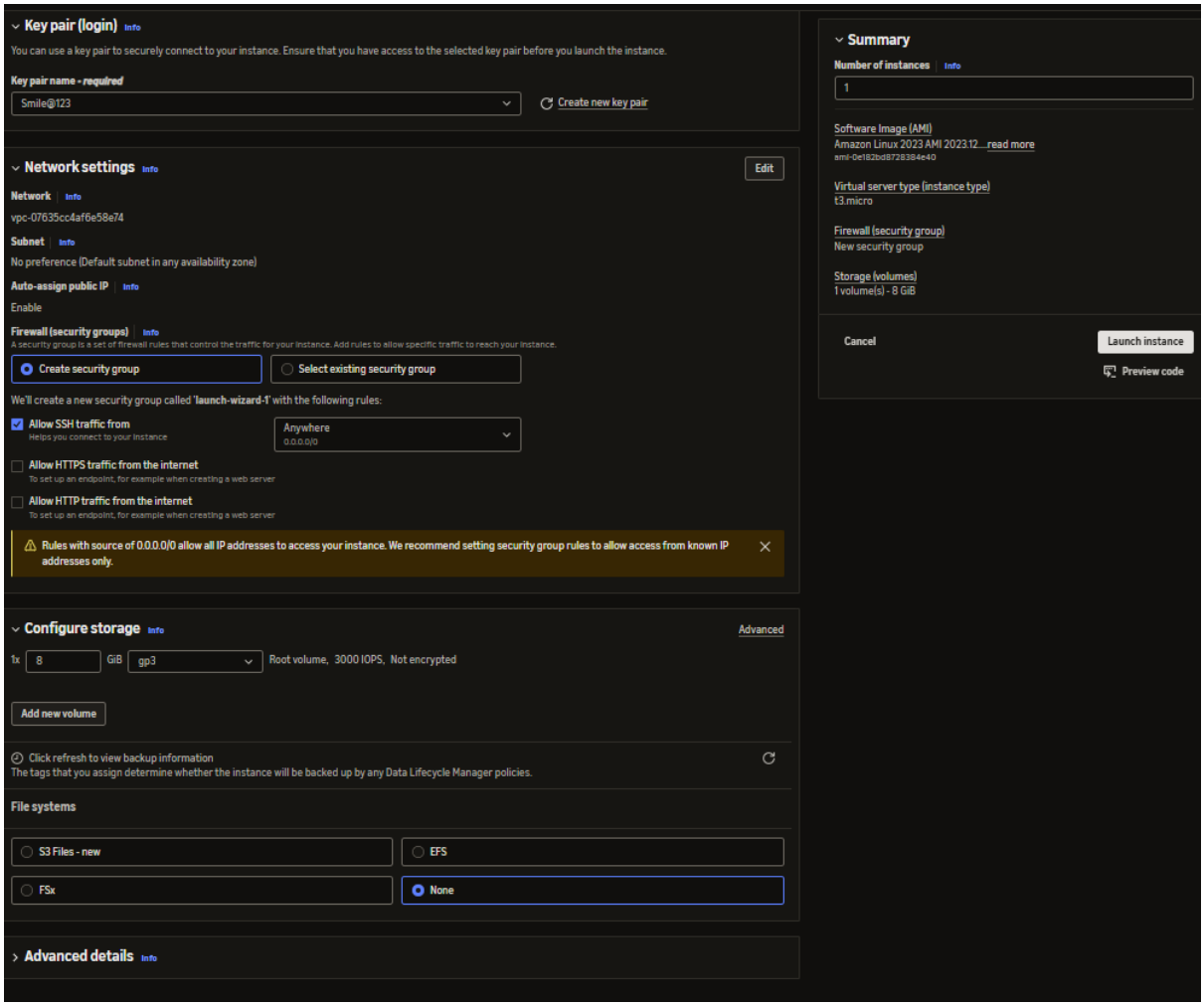
Output:

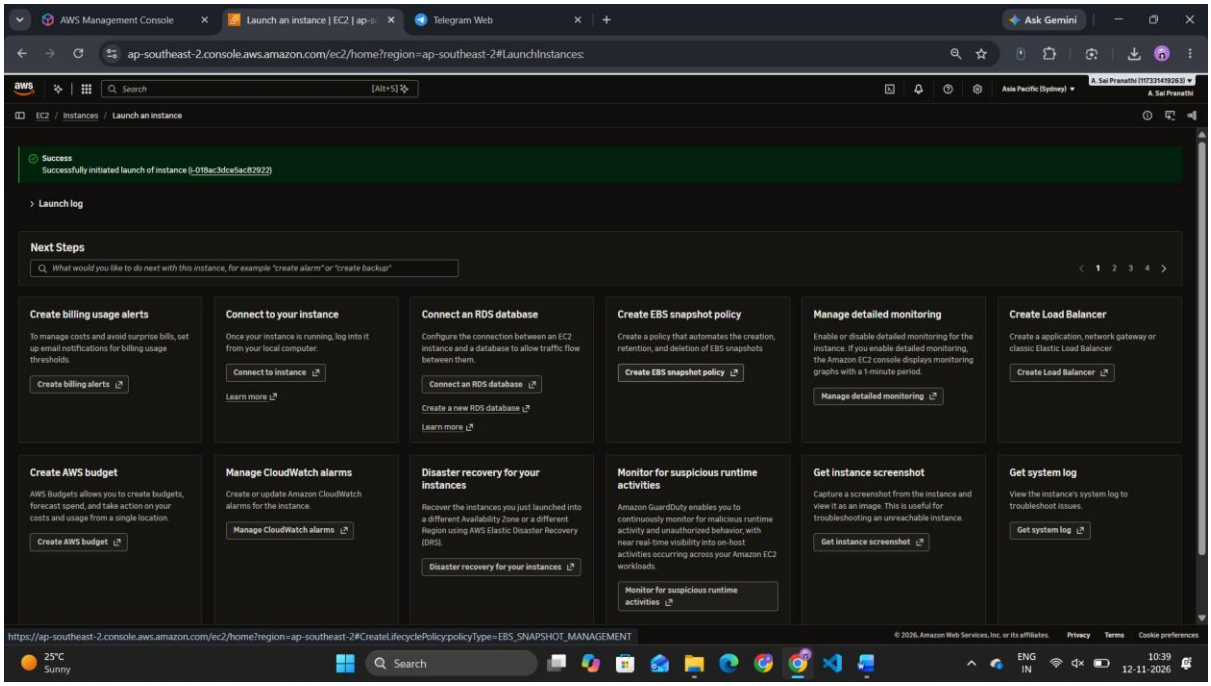



AWS Management Console
Dashboard | EC2 | ap-southeast
Telegram Web
Ask Gemini
ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#Home:
Search
[Alt+S]
Asia Pacific (Sydney)
A. Sai Pranathi (117331419263)
A. Sai Pranathi
EC2
Dashboard
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Resources
You are using the following Amazon EC2 resources in the Asia Pacific (Sydney) Region:
Instances (running)
Auto Scaling Groups
Capacity Reservations
Dedicated Hosts
Elastic IPs
Instances
Key pairs
Load balancers
Placement groups
Security groups
Snapshots
Volumes
Account attributes
Settings
Data protection and security
Allowed AMIs
Zones
EC2 Serial Console
Default credit specification
Managed resources
EC2 console preferences
Launch instance
To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.
Launch instance
Migrate a server
Note: Your instances will launch in the Asia Pacific (Sydney) Region
Service health
AWS Health Dashboard
Zones
Zone name
Zone ID
Enable additional Zones
Explore AWS
Enable Best Price-Performance with AWS Graviton2
AWS Graviton2 powered EC2 instances enable up to 40% better price performance for a broad spectrum of cloud workloads. Learn more
10 Things You Can Do Today to Reduce AWS Costs
Explore how to effectively manage your AWS costs without compromising on performance or capacity. Learn more
Introducing Spot Blueprints, a Real-Time Template Generator
Take advantage of Spot Blueprints to quickly generate CloudFormation and Terraform templates like Kubernetes and Apache Spark. Learn more
Instance alarms
View in CloudWatch
0 in alarm
0 OK
0 insufficient data
Instances in alarm
https://ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#LaunchInstances:
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
25°C
Sunny
Search
ENG
IN
10:25
12-11-2026

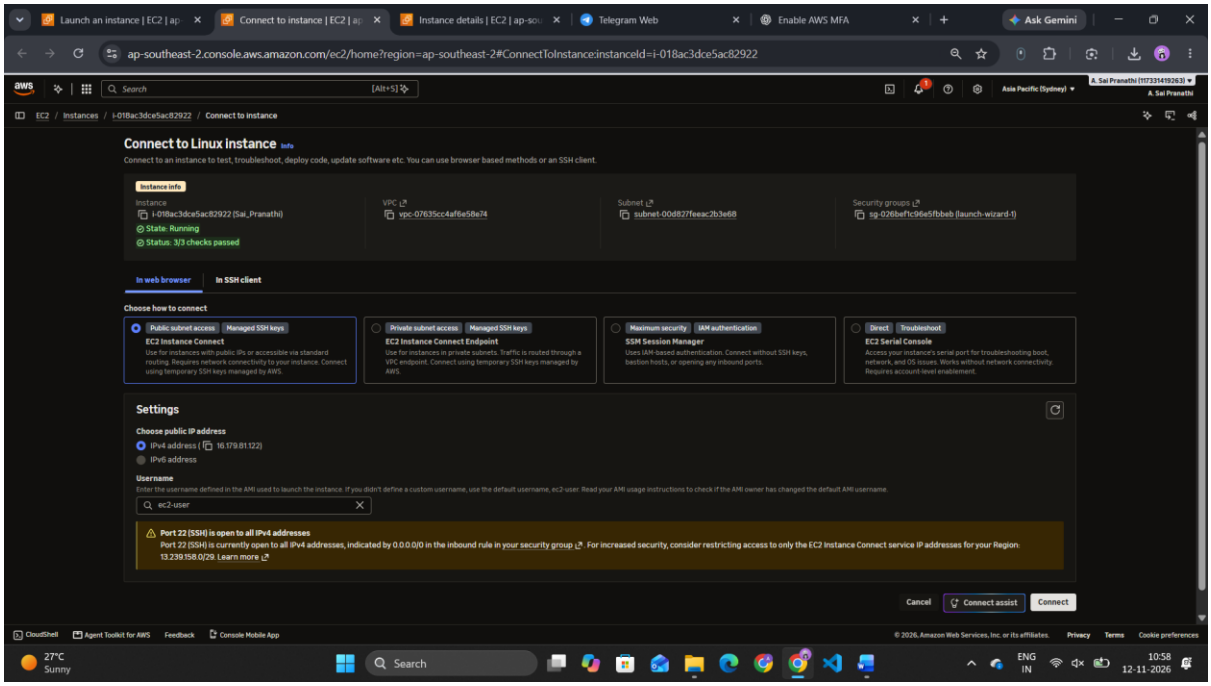

AWS Management Console
Launch an instance | EC2 | ap-s
Telegram Web
Ask Gemini
ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#LaunchInstances:
EC2 / Instances / Launch an instance
It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices
Take a walkthrough
Do not show me this message again
Launch an instance
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags
Name
Sai_Pranathi
Add additional tags
Application and OS Images (Amazon Machine Image)
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
Free tier eligible
Description
Amazon Linux 2023 (kernel-6.18) is a modern, general purpose Linux-based OS that will be supported until June 2029. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.
Amazon Linux 2023 AMI 2023.12.20261109.0 x86_64 HVM kernel-6.18
Architecture
Boot mode
AMI ID
Creation Date
Username
64-bit (x86)
uefi-preferred
2026-11-04
ec2-user
Verified provider
Summary
Number of instances
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12... read more
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
25°C
Sunny
Search
ENG
IN
10:29
12-11-2026

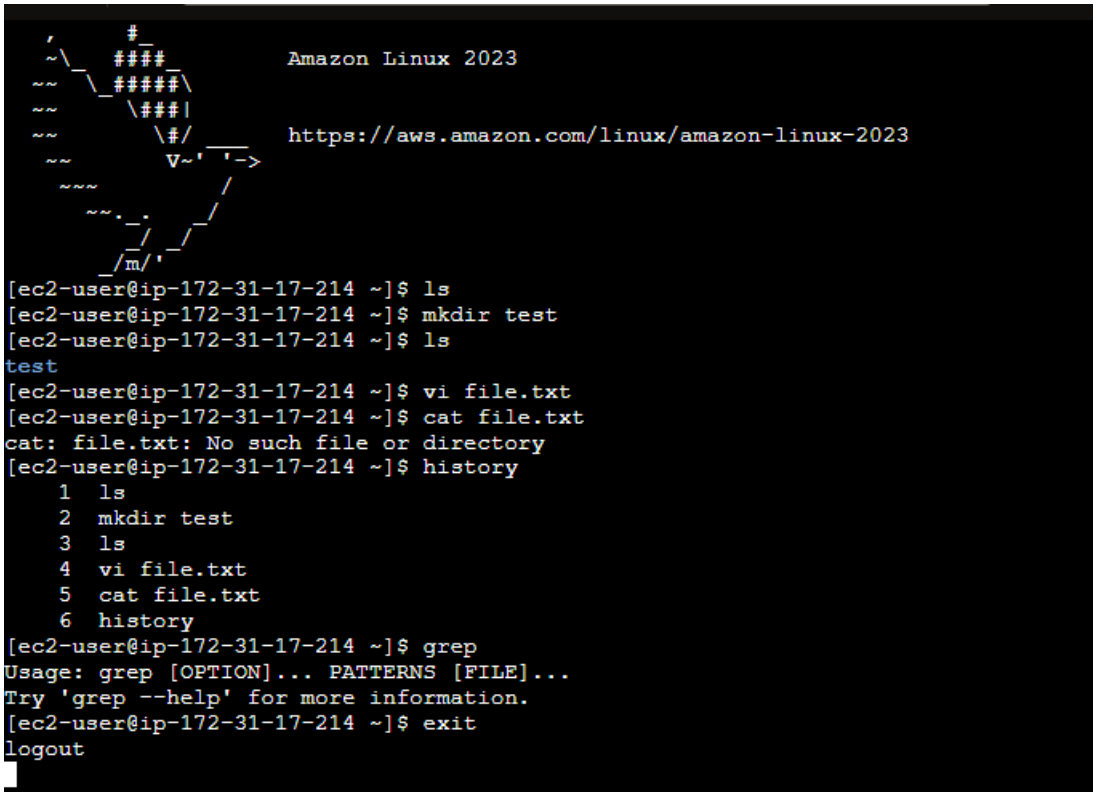

AWS Management Console
Launch an instance | EC2 | ap-s
Telegram Web
Ask Gemini
ap-southeast-2.console.aws.amazon.com/ec2/home?region=ap-southeast-2#LaunchInstances:
EC2 / Instances / Launch an instance
Instance type
Summary
Create key pair
Key pair name
Smile@123
Key pair type
RSA
RSA encrypted private and public key pair
ED25519
ED25519 encrypted private and public key pair
Private key file format
.pem
For use with OpenSSH
.ppk
For use with PuTTY
When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more
Cancel
Create key pair
Key pair (login)
Network settings
Launch instance
Preview code
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
Privacy
Terms
Cookie preferences
25°C
Sunny
Search
ENG
IN
10:32
12-11-2026

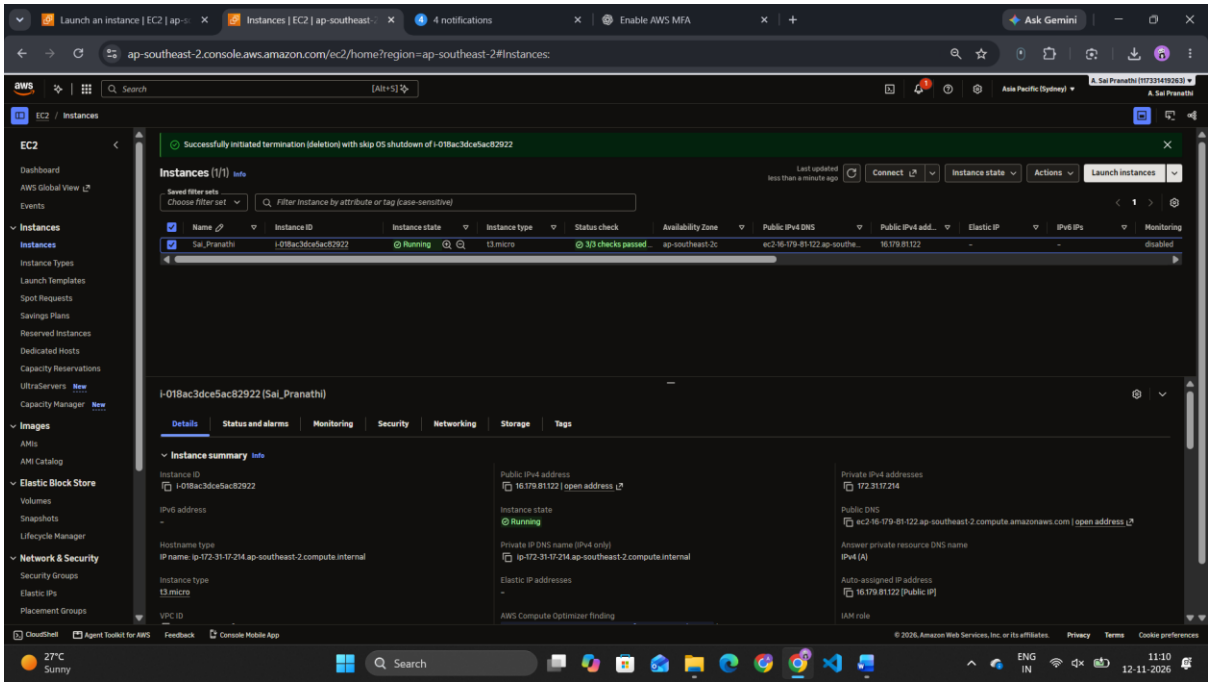

Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
Smile@123
Create new key pair
Network settings Info
Edit
Network Info
vpc-07635cc4af6e58e74
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Enable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
We'll create a new security group called 'launch-wizard-1' with the following rules:
Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server
Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.
Configure storage Info
Advanced
1x
8
GiB
gp3
Root volume, 3000 IOPS, Not encrypted
Add new volume
Click refresh to view backup information
The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.
File systems
S3 Files - new
EFS
FSx
None
Advanced details Info
Summary
Number of instances Info
1
Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...read more
ami-0e182bd8728384e40
Virtual server type (instance type)
t3.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Cancel
Launch instance
Preview code


Success
Successfully initiated launch of instance (i-018ac3dce5ac82922)
Launch log
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy
Manage detailed monitoring
Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period.
Manage detailed monitoring
Create Load Balancer
Create a application, network gateway or classic Elastic Load Balancer
Create Load Balancer
Create AWS budget
AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.
Create AWS budget
Manage CloudWatch alarms
Create or update Amazon CloudWatch alarms for the instance.
Manage CloudWatch alarms
Disaster recovery for your instances
Recover the instances you just launched into a different Availability Zone or a different Region using AWS Elastic Disaster Recovery (DRS).
Disaster recovery for your instances
Monitor for suspicious runtime activities
Amazon GuardDuty enables you to continuously monitor for malicious runtime activity and unauthorized behavior, with near real-time visibility into on-host activities occurring across your Amazon EC2 workloads.
Monitor for suspicious runtime activities
Get instance screenshot
Capture a screenshot from the instance and view it as an image. This is useful for troubleshooting an unreachable instance.
Get instance screenshot
Get system log
View the instance's system log to troubleshoot issues.
Get system log


Instance summary for i-018ac3dce5ac82922 (Sai_Pranathi)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance details
Host and placement group
Capacity reservation


Connect to Linux instance
Connect to an instance to test, troubleshoot, deploy code, update software etc. You can use browser based methods or an SSH client.
Instance info
Instance
i-018ac3dce5ac82922 (Sai_Pranathi)
State: Running
Status: 3/3 checks passed
VPC
vpc-07635cc4af6e58e74
Subnet
subnet-00d827feeac2b3e68
Security groups
sg-026befc96e5fbbeb (launch-wizard-1)
In web browser
In SSH client
Choose how to connect
EC2 Instance Connect
EC2 Instance Connect Endpoint
SSM Session Manager
EC2 Serial Console
Settings
Choose public IP address
IPv4 address (16.179.81.122)
IPv6 address
Username
ec2-user
Port 22 (SSH) is open to all IPv4 addresses
Cancel
Connect assist
Connect


Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-17-214 ~]$ ls
[ec2-user@ip-172-31-17-214 ~]$ mkdir test
[ec2-user@ip-172-31-17-214 ~]$ ls
test
[ec2-user@ip-172-31-17-214 ~]$ vi file.txt
[ec2-user@ip-172-31-17-214 ~]$ cat file.txt
cat: file.txt: No such file or directory
[ec2-user@ip-172-31-17-214 ~]$ history
1 ls
2 mkdir test
3 ls
4 vi file.txt
5 cat file.txt
6 history
[ec2-user@ip-172-31-17-214 ~]$ grep
Usage: grep [OPTION]... PATTERNS [FILE]...
Try 'grep --help' for more information.
[ec2-user@ip-172-31-17-214 ~]$ exit
logout



AIM:2C) S3 Bucket Creation and Upload Objects
I. Create an S3 Bucket for file storage.
II. Upload 5 objects with different file extensions.

Description:

Amazon Simple Storage Service (Amazon S3) is an object storage service provided by AWS. It allows users to store and manage files in containers called buckets. A bucket is created with

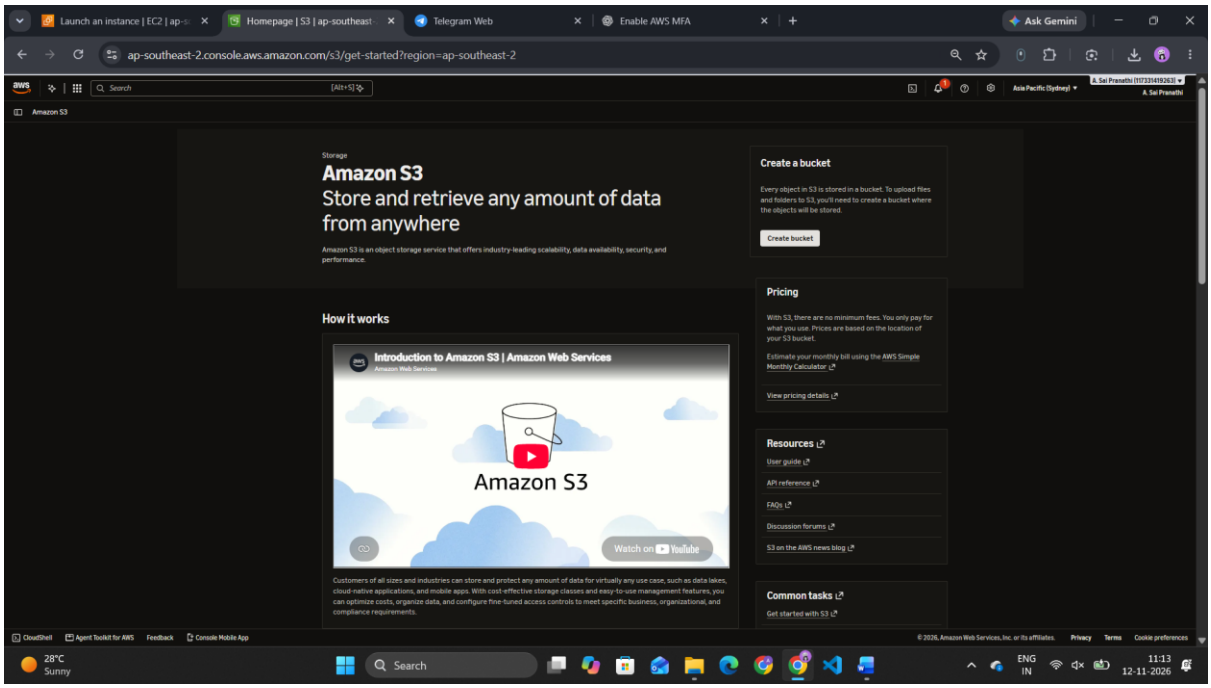
a globally unique name and an AWS Region, and it can store documents, images, videos and other types of files.

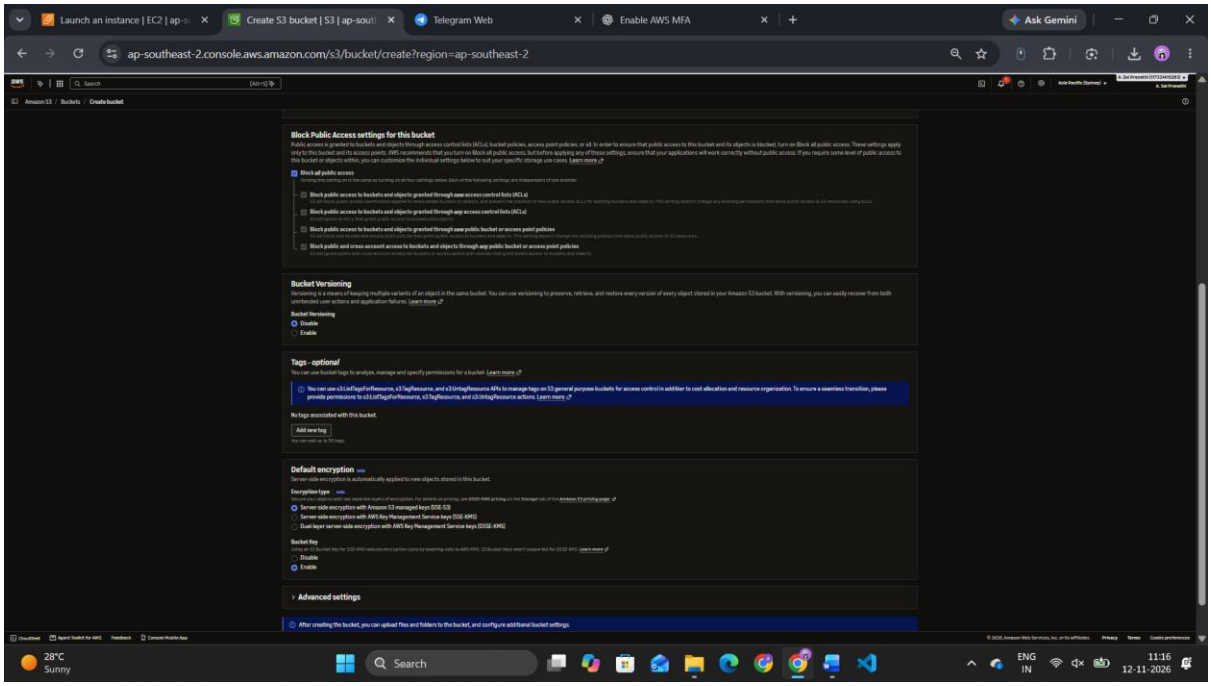
After creating the bucket, users can upload objects in different file formats. In this experiment, five files with different extensions, such as PDF, JPG, PNG, TXT and DOCX, are uploaded. The uploaded objects can be viewed and managed from the S3 bucket.

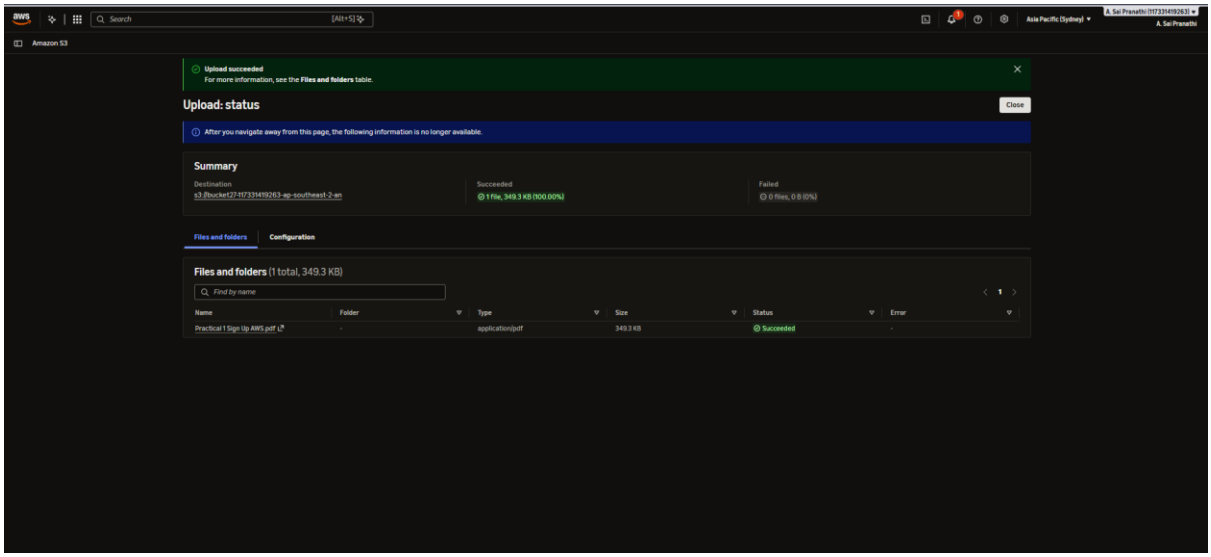
Steps:

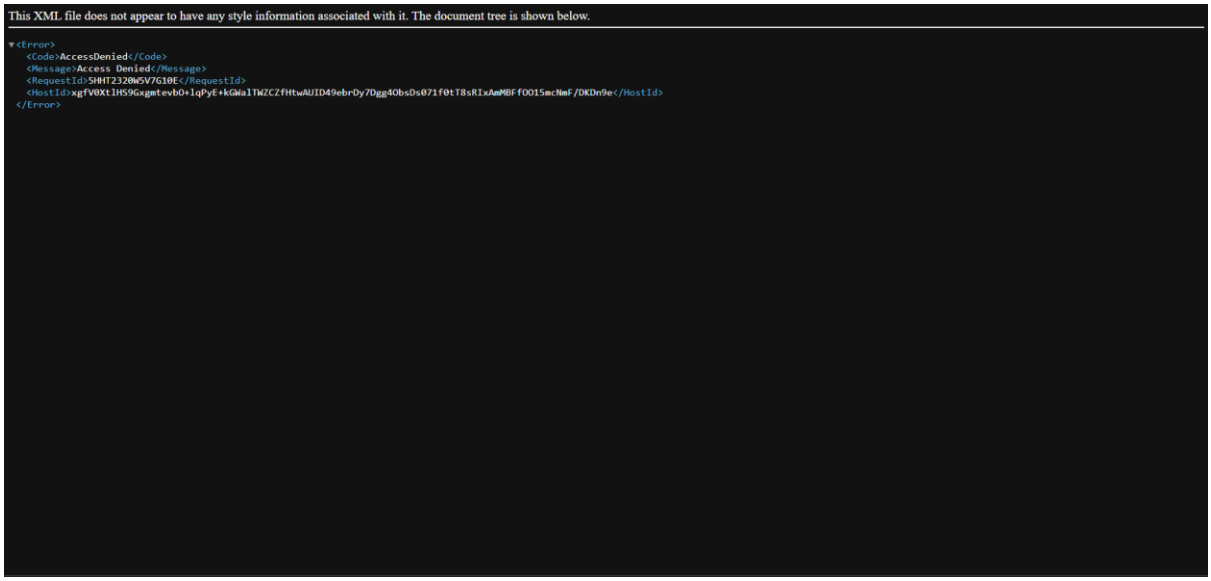
Sign in to the AWS Management Console and open the Amazon S3 service.
2. Click Create Bucket and enter a unique bucket name.
3. Select the AWS Region and keep the default settings as required.
4. Click Create Bucket to create the storage bucket.
5. Open the created bucket and click Upload.
6. Click Add Files and select five files with different extensions, such as .pdf, .jpg, .txt, .docx, and .png.
7. Click Upload and wait until all five objects are successfully uploaded.

Output:




Launch an instance | EC2 | ap-s
Create S3 bucket | S3 | ap-sout
Telegram Web
Enable AWS MFA
Ask Gemini
ap-southeast-2.console.aws.amazon.com/s3/bucket/create?region=ap-southeast-2
Amazon S3 / Buckets / Create bucket
Block Public Access settings for this bucket
Block all public access
Bucket Versioning
Disable
Enable
Tags - optional
No tags associated with this bucket.
Add new tag
Default encryption
Server-side encryption with Amazon S3 managed keys (SSE-S3)
Server-side encryption with AWS Key Management Service keys (SSE-KMS)
Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)
Bucket Key
Disable
Enable
Advanced settings
After creating the bucket, you can upload files and folders to the bucket, and configure additional bucket settings.
28°C
Sunny
Search
ENG
IN
11:16
12-11-2026


Amazon S3
Asia Pacific (Sydney)
A. Sai Pranathi
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://bucket27-417331419263-ap-southeast-2-an
Succeeded
1 file, 349.3 KB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 349.3 KB)
Find by name
Name
Folder
Type
Size
Status
Error
Practical 1 Sign Up AWS.pdf
application/pdf
349.3 KB
Succeeded


This XML file does not appear to have any style information associated with it. The document tree is shown below.
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<Code>AccessDenied</Code>
<Message>Access Denied</Message>
<RequestId>5HHT2320W5V7G10E</RequestId>
<HostId>xgfV0XtlHS9GxgmtevbO+lqPyE+kGWalTWZCZfHttwAUID49ebrDy7Dgg4ObsDs071f0tT8sRIxAmMBFfOO15mcNmF/DKDn9e</HostId>
</Error>